SmartFlow : A Database Driven Intelligent Traffic Management System

Literature Survey

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Title: Overview of Road Traffic Management Solutions based on IoT and AI

Author: Asma Ait Quallane, Ayoub Bahnasse, Assia Bakali, Mohamed Talea

Year: 2021

Objective: To minimize road traffic congestion through intelligent and efficient traffic management using technologies such as AI, IoT, and Big Data.

Methodology: Reviewed and classified existing research into routing, traffic light control, and network traffic management, then compared their technologies, performance, and limitations.

Results: AI, IoT, and Big Data significantly improve traffic management, but many systems still lack emergency vehicle prioritization and coordinated intersection control.

Limitations: Most approaches focus on isolated intersections, require expensive infrastructure or widespread connected vehicles, and often do not prioritize emergency vehicles.

Relevance to Our Project: This paper is relevant to our project because it discusses AI-based traffic management solutions and their limitations. Unlike many existing systems, our project uses a **single rotating camera** to reduce infrastructure costs and provides **priority to emergency vehicles and buses**, enabling more efficient and intelligent traffic signal management.

Title: A Review of the Self-Adaptive Traffic Signal Control System Based on Future Traffic Environment

Author: Yizhe Wang, Xiaoguang Yang, Hailun Liang, Yangdong Liu

Year: 2018

Objective: To improve adaptive traffic signal control systems that can respond to real-time traffic conditions and reduce congestion

Methodology: The paper reviews different generations of adabtive traffic signal control and investigate data-driven approaches such as fuzzy logic, genetic algorithms, neural network and multi-agent reinforcement learning.

Data: Real-time traffic information such as vehicle position, speed, queue length, stopping time, traffic flow, and data from sensors, video

Result: Reinforcement learning based control provides real time, self learning and model free adaptive control and improve traffic efficiency compared with conventional fixed time.

Limitations: Many existing approaches depend on complex infrastructure, large amounts of traffic data and precise traffic models. Multi intersection control was still considered an emerging research area.

Relevance to Our Project: Our project similarly uses real-time traffic information to dynamically adjust traffic signals to reduce congestion. However, our system uses a single camera for vehicle detection, which can reduce infrastructure and sensor costs. We also prioritize emergency vehicles and buses, extending the adaptive control system to prioritize important vehicle types.

Title: The Role of Intelligent Transport Systems and Smart Technologies in Urban Traffic Management in Polish Smart Cities

Author: Yizhe Wang, Xiaoguang Yang, Hailun Liang

Year: 2025

Objective: Improve urban traffic management by using AI and IoT to reduce congestion, emissions, travel time and improve safety.

Methodology: AI, IoT, camera, sensors, machine learning, smart traffic lights. GIS, big-data analytics and ML are then used to analyze and predict traffic and evaluate the system.

Data: Traffic data is collected using GPS, smartphones, cameras, IoT sensors and municipal reports.

Results: Intelligent traffic system improved traffic flow, reduced waiting, congestion and improved safety in several Polish cities.

Limitations: Requires significant infrastructure, multiple data sources, high investment, privacy and cybersecurity considerations.

Relevance to Our Project: Our project applies the same concept of AI-based real time traffic monitoring and dynamic signal control, but uses a single camera, reducing infrastructure requirements while giving priority to emergency vehicles and buses

Title: Optimal deep neural network based road traffic management system for Internet of Things based smart city environment

Year: 2026

Author: Khaled Abdullah Almejalli

Objective: To develop IoT based decision support system for efficient road traffic management by analysing real time traffic data, predicting traffic conditions and reducing congestion.

Methodology: The proposed HFSDNN-DSSRTM model performs data preprocessing using missing-value handling and Min-Max normalization. It then uses hybrid feature selection methods like MI, Fisher Score, SFS, RFE and Random Forest Importance to select important features. Finally, a Temporal Convolutional Network with Attention Mechanism **is** used for traffic classification and prediction.

Data: The model was tested using a Smart Traffic Management dataset containing 2,000 instances and 12 features**.**

Result: Using an 80:20 train-test split, the model achieved **98.75% accuracy**, with precision, recall and F1-score also showing high performance.

Limitations: The study relies on a relatively small dataset of **2,000 instances** and depends on labelled traffic data. The authors also mention limitations regarding scalability, interpretability, heterogeneous data integration and real-world applicability.

Relevance to Our Project: The paper is relevant because it demonstrates how AI and real-time traffic data can be used to reduce congestion and dynamically manage traffic signals. Our project similarly aims to reduce road congestion using a single camera**.** Our system also differs by giving priority to emergency vehicles and buses, making it more focused on real-time vehicle detection and signal optimization.

Title: Real-Time Vehicle Detection Based on Improved YOLO v5

Year: 2022

Author: Yu Zhang, Zhongyin Guo, Jianqing Wu, Yuan Tian, Haotian Tang, Xinming Guo

Objective: To improve vehicle detection accuracy in different highway traffic scenarios, particularly for small vehicles, and reduce false detections.

Methodology: The author developed a multi-scenario vehicle dataset containing 2844 images and trained an improved YOLO v5 model. They introduced Flip-Mosaic data augmentation, replaces GIou loss with CIou loss and used adaptive calculation to improve detection of small vehicles.

Data: 2844 images from highway surveillance videos and the BIT Vehicle Dataset. Vehicles were classified into 8 categories such as Bus, Minibus, Sedan, Taxi, Heavy Truck, Truck, SUV and Special Vehicle.

Results: The improved YOLO v5 showed better vehicle detection performance than standard YOLO v5, particularly for small vehicles.

Limitations: The dataset was limited because obtaining highway surveillance videos was difficult. More scenes, camera angles, lighting conditions and bad-weather data were needed to improve generalization.

Relevance to Our Project: Our project also uses camera based vehicle detection for intelligent traffic management. The shows how YOLO-based detection can identify different vehicles in real-time traffic environments. Our project differs by using a single camera to reduce infrastructure and by using vehicle detection to support adaptive traffic signal control with priority given to emergency vehicles and buses.

Title: Revolutionizing Urban Mobility: A Systematic Review of AI, IoT, and Predictive Analytics in Adaptive Traffic Control Systems for Road Networks

Year: 2025

Author: Carmen Gheorghe, Adrian Soica

Objective: To review how AI, IoT and predictive analytics can be integrated together into adaptive traffic control system to reduce traffic flow, congestions and improve safety and support emergency vehicle priority.

Methodology: A literature review was conducted using Scopus and Web of science. Studies from 2020-2024 were collected and filtered and **144** relevant papers were finally selected for detailed review.

Key Findings: AI based traffic model can dynamically adjust signal timing according to traffic conditions, IoT sensors and cameras provide real time traffic information and Predictive Analytics can forcast traffic congestions and help optimize traffic signals.

Results: The reviewed studied reported improvements in traffic efficiency and congestion reduction. One DRL-based approach reported up to 31% improvement compared with traditional pre-defined signal timing.

Limitations: High infrastructure cost, data privacy and security issues, reliability on real-time data and difficulty integrating different technologies.

Relevance to Our Project: Our project applies the same concept of AI-based adaptive traffic signal control using a single camera to detect and classify vehicles instead of expensive sensors. The system can dynamically adjust signal timings and prioritize emergency vehicles and buses, helping reduce congestion and improve traffic flow while lowering infrastructure cost.

Title: Artificial Intelligence-Based Real-Time Traffic Management

Author: Abbas Kadkhodayi, Mohammad Jabeli, Hassan Aghdam, Shahin Mirbakhsh

Year: 2023

Objective: To reduce urban traffic congestion and improve traffic routing by using Ant Colony Optimization(ACO) within a multi-agent architecture supported by IoT.

Methodology: The system represents the road network as a graph consisting of intersections and roads. Multiple computational vehicle agents communicate with intersection controller agents. Ant Colony Optimization is used to find efficient routes, while IoT based data provides real time information such as traffic flow, density, occupancy and accidents.

Results: The study indicated that multi agent traffic control can outperform conventional traffic controllers. ACO can help identify efficient routes, while real time communication between agents enables better traffic management and emergency routing.

Limitations: The proposed system relies on a distributed multi agent architecture and IoT based communication which can increase system complexity and infrastructure requirements. ACO based approach focuses mainly on optimal path routing rather than detecting vehicles and dynamically controlling traffic signal.

Relevance to Our Project: Our SmartFlow project also uses Ai and real time traffic information to improve traffic signal management. However, our system focuses on camera based vehicle detection and dynamic traffic signal control, whereas this paper primarily focuses on ACO based path routing using distributed agents and IoT data.

Title: A Lightweight Edge AI Framework for Adaptive Traffic Signal Control in Mid-Sized Philippine Cities

Author: Alex L.Maureal, Franch Maverick A. Lorilla, Ginno L. Andres

Year: 2023

Objective: To reduce urban traffic congestion using ACO algorithm within a distributed multi agent system using Iot technology for efficient path routing and traffic management.

Methodology: Tis system models the road as a graph where intersections are noted and roads are edges. IoT enabled vehicles share traffic information with intersection controller agents. A multi-agent system manages the communications with intersections and ACO algo determines the optimal traffic route.

Result: The study concludes that combining IoT, Multi agent systems and ACO can improve traffic routing, reduce congestion, minimize travel time and lower accident risks.

Limitations: The work is mainly conceptual with limited real world validation. Requires extensive IoT infrastructure and reliable communication between agents. Scalability under extremely large traffic networks is not experimentally evaluated.

Relevance to Our Project: This paper is closely related to our smart traffic management system, but instead of distributed multi-agent ACO system, our project uses a single camera YOLO based vehicle detection system, making it more cost effective and our system gives priority to emergency vehicles and buses.

Title: IoT-Based Smart Traffic Management System using Computer Vision and Deep Learning

Author: Liung Wing

Year: 2026

Objective: To develop an IoT based smart traffic management system that integrates computer vision and deep learning for real time traffic monitoring and adaptive signal control

Methodology: IoT cameras and sensors continuously collect traffic data. Computer vision models like CNN, YOLO etc detect and classify vehicles and estimate traffic density. The reinforcement learning adaptively controls traffic signals.

Data: Continuous real time video streams from cameras, IoT sensor data and diverse vehicle image dataset for training computer vision models.

Results: Vehicle detection accuracy reached 92% using YOLOv4. Adaptive signal control reduced average vehicle waiting time by 30%**,** increased traffic throughput by 25%**,** and reduced decision latency to under 100 ms through edge computing.

Limitations: High deployment and maintenance cost. Depends on high quality camera and sensor coverage. Performance decreases in poor weather and low light conditions. Privacy concerns due to continuous surveillance

Relevance to Our Project: This paper closely align with our Smart Traffic Management System. It uses IoT, computer vision, YOLO and deep learning for adaptive traffic control. Our project similarly employs YOLO based vehicle detection but focuses on cost effective implementation.

Title: Real-time traffic monitoring system using IoT-aided robotics and deep learning techniques

Author: Mohammed Qader Kheder, Aree Ali Mohammed

Year: 2024

Objective: To develop IoT-based real time road monitoring system that collects traffic data using sensors and cameras, detect traffic lights and provides real time road information to users to improve traffic flow and safety.

Methodology: The system combines IoT sensors and computer vision. An Autonomous Robotic Car uses cameras, IR sensors, ultrasonic sensors and GPS to collect road data. Inception-V3 with transfer learning is used for traffic light detection.

Data: GTSRB, EGTSRB, for traffic sign recognition and LISA for traffic light detection

Results: LeNet-5 achieved 99.12% on GTSRB and 99.78% on EGTSRB. Inception-V3 achieved 98.6% on LISA.

Limitations: Requires Wi-Fi for real time notifications. Performance can be affected by bad weather. ARC operated only on reads with black markings. Only 43 traffic sign classes are supported.

Relevance to Our Project: The paper supports the use of camera based traffic monitoring and AI for real time traffic management**.** Your project differs by focusing on traffic-density-based signal optimization, using a single camera to reduce infrastructure cost, while also prioritizing emergency vehicles and buses**.**